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Understanding fans' responses to sustainability: a segmentation of sport event goers' pro-environmental intentions and behaviours

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ABSTRACT

It is widely argued that sport can be a tool to market environmental sustainability both through and within it, yet there exists a notable gap between individuals' environmentally friendly interests and actions in and around sport. In this study, we centre on sport event goers in an attempt to better study and segment them, based on their pro-environmental intentions and behaviours, in order to better identify how future marketing communication campaigns on environmental sustainability could be better tailored and targeted to each segment. Quantitative data collected through an online survey during a football match ($n = 578$) are analysed for patterns to be identified, before a cluster analysis is conducted. Three distinct clusters are identified, each pointing to a different form of the value-action gap. The segmentation profiling obtained through this study illustrates the differences among individuals that need to be understood first, before environmental sustainability campaigns are to be created.

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Introduction

In the ever-developing research focusing on sport ecology, it is highlighted that sport has a bidirectional relationship with the environment, in which sport can both affect and be affected by the environment (Trail & McCullough, 2020). In terms of the former, sports have been found to have negative effects on various aspects of the environment, such as air, soil, noise and water pollution, and depletion of natural resources for energy production (Mallen & Chard, 2011). In the case of sport events, in particular, it is noted that they can have significant negative effects on the environment. For a case in point, Pereira et al. (2017) reported that the carbon footprint of the 2016 World Cup in Brazil, which lasted less than a month, generated 2.7 million tons of greenhouse

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gases, which they argue corresponds to the production of a country the size of Malta for a period of 1 year. According to the researchers, transportation (83.7%), facilities (9.65%) and accommodation (5.7%) were the main contributors to this production. Through this, it appears that transport to sport can be the main burden on the environment, something that numerous studies have also argued (see Dolf & Teehan, 2015; Triantafyllidis et al., 2018). This travelling and the corresponding choice of how sport fans travel to a sport event have in fact sparked a wider line of research, with multiple studies focusing on analysing fans' travelling behaviours in the context of sports (Casper & Pfahl 2012a; 2012b; Casper et al., 2017, 2020, 2023; Loewen & Wicker, 2021; McCullough & Cunningham, 2011; McCullough, 2013).

Within these studies, a notable gap is highlighted between fans' intentions and potential interest in adopting sustainable practices and their actual behaviours which more often than not deviate from them (e.g. in Casper & Pfahl 2012a; 2012b; Loewen & Wicker, 2021; McCullough & Cunningham, 2011; Wicker, 2018, 2019). This notable gap has created further challenges for sport organisations, which in response have engaged at times in sophisticated campaigns focusing on environmental sustainability and asking for stakeholder engagement not only in an effort to reduce the carbon footprint of the sport events organised but also in an attempt to assist in the wider efforts to increase environmentally sustainable practices within their fan base (McCullough et al., 2016; Trail & McCullough, 2021). One of the many examples of these efforts to market sustainability through sport is the recent partnership between UEFA and the European Commission, which has collaboratively created a campaign promoting climate action among fans and thus used football's platform to raise public awareness about Europe's climate and energy priorities (UEFA, 2022).

Such efforts have led the authors to argue that the sports industry's biggest contribution to climate action can be to provide its platform to engage with its millions of fans and participants, and that is why they introduced the concept of Scope F emissions 'attributable to the influence that a football club or another sporting organisation or athlete has on fans' (Atkins et al., 2022). Through this, it is therefore suggested that apart from the "traditional" approach of reducing in-house Scope 1,2,3 emissions, sports organisations can multiply their positive contribution by engaging their spectators to environmental sustainability initiatives. Marketing sustainability in and through sport has thus become a common practice in the world of sport; however, these marketing communication campaigns, while praised for their design and efforts to improve over time (Cayolla et al., 2023; Trail & McCullough, 2020, 2021), seem to miss out on a fundamental element of modern-day marketing, that of customisation for the audience. Despite wide arguments on the disparities among sport stakeholders and their sport consumption habits (Fuchs et al., 2024), existing studies on the gap between individuals' pro-environmental sustainability intentions and behaviour, and any corresponding campaigns to combat it, continue to adopt a rather blanket, one-size-fits-all approach for all fans. In line with wider efforts in strategic marketing, within and beyond sport (Manoli, 2022), one could question whether a more tailored approach might be more beneficial for closing this gap, by first better understanding the individuals in question.

In this study, we aim to do so, by studying sport event goers and segmenting them based on their pro-environmental intentions and behaviours, in order to better identify how future marketing communication campaigns on environmental sustainability could

be better tailored and targeted to each segment. Through this study, insights are offered on the different forms that the value-action gap can take, thus responding to a much more pressing need identified within existing literature and practice (Herold et al., 2024).

Literature review

A number of studies have investigated the environmental behaviours of individuals in the context of sports (Casper & Pfahl 2012a; 2012b; Casper et al., 2017, 2020, 2023; Loewen & Wicker, 2021; McCullough, 2013; McCullough & Cunningham, 2011). Most of these studies are based on the theories of planned behaviour (Ajzen, 1991), reasoned action (Ajzen & Fishbein, 1980), and the Value Belief Norm framework (Stern, 2000). Even though in some of these studies environmental consciousness has been found to be positively associated with pro-environmental behaviours (Casper et al., 2023; Thormann & Wicker, 2021; Wicker, 2019), in others, the environmental concern did not lead to pro-environmental actions (Casper & Pfahl 2012a; 2012b; McCullough, 2013; McCullough & Cunningham, 2011; Wicker, 2018, 2019). These results suggest that environmental concerns do not automatically lead to pro-environmental actions, pointing to the need for further examination of individuals involved in sport.

As such, several constraints have been identified as inhibiting sport participants and fans/spectators from acting in pro-environmental ways, such as convenience and accessibility (McCullough, 2013), lack of sufficient knowledge (Casper & Pfahl 2012a), cost of travel (Loewen & Wicker, 2021), and suspicion of greenwashing (McCullough, 2013). Specifically, Loewen and Wicker (2021) concluded that environmentally conscious fans travelling to Bundesliga matches did not generate a lower carbon footprint, supporting the existence of a 'value-action' gap, as described by Blake (1999), suggesting that fans' concern about the environment did not lead them to actually use more sustainable travel modes to attend Bundesliga matches. Similar arguments were made in studies conducted by Casper & Pfahl (2012a, 2012b), Wicker (2018), Wicker (2019), and McCullough and Cunningham (2011), with this gap consistently identified, and the need for further study on the individuals and this gap is highlighted.

Further research on the environmental value-action gap has argued that it can be explained by the low-cost hypothesis (Diekmann & Preisendörfer, 2003), according to which environmental consciousness will result in pro-environmental behaviours when the cost related to such behaviours is low. On the contrary, it is argued that when the cost in terms of money, time, and convenience related to environmentally friendly behaviours is too high, people will not act based on environmental consciousness. Based on this hypothesis and considering that travel to sport events is relatively high-cost, fans' environmental concerns will not lead them to behave in pro-environmental manners (Loewen & Wicker, 2021). Interestingly and not unexpectedly, Loewen and Wicker (2021) also found that fans who were strongly committed to their team produced a greater amount of carbon emissions, since going more often to a sport event would subsequently cause a higher carbon footprint (Diekmann & Preisendörfer, 2003). Regarding transportation to a sport event, fans' travel behaviours have been investigated by Dosumu et al. (2017), whose study highlighted that travelling to and from a football match can have a considerable impact on the environment, with fans nevertheless favouring the car in their choice of mode of transport. Convenience and saving time were regarded as the

main reasons for choosing the car, even for matches played in the stadium of the opposing team. However, in the context of Austrian football, Herold et al. (2024) found that 52,8% of Rapid Vienna's fans utilised the public transport system while over 40% came by car and only a minor portion (4,6%) walked, cycled, or used a coach to go to the stadium.

While exploring the issue of marketing sustainability through sport, Casper et al. (2023) examined how the environmental behavioural intentions of fans were affected by environmental education efforts at a sporting event seeking to promote environmental sustainability. The researchers found that the values, beliefs, and norms of respondents predicted pro-environmental intentions at sports events and everyday life (VBN – Values, Beliefs, Norms – model). Similarly, when measuring environmental values and expectations for environmental activities, at a sports event and at home environmental behaviours, it was suggested that fan engagement is affected by environmental activities in an intercollegiated sporting context (Casper et al., 2017). According to their study, understanding how fans perceive environmental actions, both emotionally and cognitively, is important in evaluating the success of environmental efforts. As such, it was argued that fans with high environmental values would expect action from their college athletic department and are likely to engage in environmental action during events and in their daily lives. In addition to this, the researchers noted that a sport organization's environmental sustainability campaign attracts fans with low identification, thus creating an opportunity to strengthen relationships with this cluster of fans. This in turn could hint on the fact that different segments of fans might respond differently to pro-environmentally responsible campaigns, once again inviting for more research on understanding them better.

A modified Value Belief Norm framework was also used to explain individuals' behavioural intentions, suggesting that fans' personal norms were positively connected to the environmental efforts of sport organizations (Casper et al., 2020). As a result, a season-long environmental campaign was found to lead fans towards adopting a more sustainable behaviour in their daily lives, suggesting that marketing sustainability through sport can have great potential (Trail & McCullough, 2020; 2021). In addition to this, an increased recognition of the investigated team's environmental efforts during the season was noted in the study, indicating that the environmental practices applied by the team may act beneficial in terms of image enhancement.

More recently, Cayolla et al. (2023) conceptualized a model linking fans' perceptions of pro-environmental activities of their club to the three dimensions of the triple bottom line (TBL), namely the environmental, social, and economic benefits to measure the impact of a club's sustainability actions. The model was applied to the fans of a professional football team, and results showed that fans' perceptions of their club's sustainability initiatives positively affected their social behaviours with regard to wider environmental issues. Results also indicated that the football team's pro-environmental actions had a positive impact on fans' pro-environmental behaviours in their daily lives, for example, regarding recycling and conserving water and energy at home, consuming less meat, and even when purchasing a car or a house.

This (Cayolla et al., 2023) along with and other studies (e.g. Trail & McCullough, 2020; 2021) have highlighted that sport can help market sustainability within and beyond sport, something that is also widely argued by organisations such as the UN, without

nevertheless shedding additional light on the value-action gap (Blake, 1999) and what sport organisations can do about it. At the same time, existing research has also ignored any differences that might exist among fans regarding their value–action gap. When considering the recent arguments on the differences among fans regarding the importance they place on social and environmental sustainability and how that informs their sport-related consumption when purchasing sport merchandise (Fuchs et al., 2024), one can wonder whether such differences can be noted in the gap between sport event goers' wider pro-environmental sustainability intentions and actions. This study aims to segment sport event attendees based on their pro-environmental intentions and actions, in order to better inform the tailoring of future marketing communication campaigns on environmental sustainability for each segment.

Methods

In line with the aim of this research, this study focused on a particular sport event in Greece, a context that has been largely understudied, potentially, offering novel insights. Data were collected during the Greece vs Ireland football match, for the UEFA European Qualifiers, on the 16th of June 2023, in the OPAP Arena, with the support of the Hellenic Football Federation and UEFA. The OPAP Arena, newly built home to AEK Athens FC, is situated in Nea Filadelfeia, a suburban area of Athens. It is well connected by public transportation, including buses and overground-metro services, making it accessible for spectators. The location, while not in the city centre, benefits from a public transportation system while suffering from a lack of parking space due to the dense population of the area.

A web-based survey, in google forms, was distributed at various gates of the OPAP Arena before and during the match, facilitated by a QR code displayed on stadium screens and gates, accompanied by encouraging messages broadcasted in the stadium. A total of 635 responses were acquired from which the 57 were excluded due to incomplete answers, resulting in a final sample of 578 responses.

The survey contained four key areas, including demographic information, transportation preferences, fans' everyday behaviours, and sport event behavioural intentions related to environmental sustainability. In particular, related to fan's transportation, questions related to the transportation mode and the reasons for selecting the particular mode of transport were adopted from previous studies (Triantafyllidis & Davakos, 2019; Triantafyllidis et al., 2018; Wicker, 2019). Participants were asked to define the mode(s) of transport they used to reach the stadium, the area they started their journey and the reason for selecting the particular mode of transport from a pre-defined selection list.

Regarding fans' everyday behaviours and sport event behavioural intentions related to environmental sustainability. Two items were used to examine everyday (private-sphere) and related to sport event (public-sphere) environmental behaviours (Casper et al., 2023). Fans were asked to select, from a list, which particular, environmental action they adopt at home and which of them they would be willing to adopt in the context of the sport event.

Specifically, fans responded to seven-items related to their sport event behavioural intentions including ' I would: use environmentally friendly transportation, buy merchandise made of recycled material, opt for paper cups and/or bottles instead of plastic, opt for vegetarian products, dispose my waste in the waste-separation bins, none of the above/

All of the above'. Following that, they have been asked regarding their everyday behaviours including questions such as 'I purchase sustainable products, I save energy at home and/or at work, I try to reduce my waste, I avoid food wasting, I recycle, and I do not take any action'. These items were adapted from Casper et al. (2023), describing the most common environmental behaviours in the Greek context.

The study responders were mostly male (79%) and under 25 years old (53%). Initially, data were analysed in order to identify any demographic related patterns in relation to their pro-environmental sustainability intentions or behaviours and any relations between these two, followed by a cluster analysis to detect segments in the sample, based on their intentions and actual behaviours.

Results

Results showed that regarding transportation, most fans used automobiles (55%), followed by the metro (23%) and buses (14%), to reach the venue. Interestingly, female fans demonstrated a higher frequency of car usage (64.2%) compared to male fans (52.9%), yielding an 11% difference ($p = 0.050$). Conversely, bus travel was more commonly favoured by males (16%), nearly triple the female percentage ($p = 0.00351$). Convenience emerged as the paramount factor for selecting these means of transport (66%), followed by time efficiency (30%), comfort (20%), and economic factors (19%). Notably, safety (12%), reliability (10%), and environmental consciousness (7%) appeared to be less of a concern for the participants. For these reasons, only time-saving was found to be different by gender (p-value = 0.0181) with 38.5% of females considering it as an important determinant for their selection compared with 27.1% of males. Table 1 summarises the travel preferences, environmental awareness, and actual behaviours by gender and age group.

The Wilcoxon signed-rank test revealed a significant difference in age distribution between those who prefer cars (mean age: 31.4 years) and those who prefer buses (mean age: 22.7 years). Further analysis revealed a statistically significant difference ($p < 0.05$) in mean age based on the reported priority when choosing transportation. Those prioritizing time-saving exhibited a higher mean age (30.6 years) compared to those who did not (28.7 years). Conversely, those prioritizing costs exhibited a lower mean age (25.5 years) compared to those who did not (30.2 years).

In terms of the association between environmental awareness and transportation mode selection among attendees, logistic regression models were employed to quantify the effect of considering environmental awareness as an important reason to select transportation and their final selection of the way to go to the stadium. Environmental awareness is included as an independent variable, and the selection of each mode of transportation is represented by a separate binary response variable. Non-significant variables were removed using a stepwise selection procedure based on the Akaike Information Criterion.

The findings revealed a positive association between environmental awareness and the use of public transportation. Individuals who reported environmental awareness as an important factor when choosing their mode of transit had significantly higher odds of selecting the metro compared to other options (approximately 50% increase in odds). Furthermore, environmental awareness was associated with a substantial increase in the

Table 1. Travel preferences, environmental awareness, and actual behaviours by gender and age group.

Variable	Gender	Age (mean age of Yes vs No)
Means of Travelling	Car F 63.2% (+10.2%) $p = 0.048$ Bus M 16.0% (+10.9%) $p = 0.002$ Airplane F 5.1% (+3.5%) $p = 0.031$	Car by Older (31 vs 27) p. <0.001 Bus by Younger (23 vs 30) p. <0.001
Reasons for Travel	Time saving F 38.5% (+11.4%) $p = 0.018$ No difference in other reasons incl. Safety ($p = 0.63$)	• Time-saving by Older (30.6 vs 29) $p = 0.021$ • Cost by Younger (25.5 vs 30) $p = 0.006$ • Safety: ○ NS for age difference MS when using 6 age categories (higher for under 20s and for 40-55 ages) $p = 0.78$
Environmental Awareness/Desire to participate in the following actions	• Buy recycled products F 53% (+16%) $p = 0.025$ • Paper cups/bottles F 65% (+15%) $p = 0.002$ • Vegetarian products F 30% (+14%) $p < 0.001$ • Waste-separation bins F 75% (+20%) $p = 0.001$ • No-engagement M 37% (+3.7%) $p < 0.001$ • Full-engagement F 24% (+10%) $p = 0007$	• Environmentally friendly transportation by older (30 vs 28) $p = 0.005$ • Paper cups/bottles Older (32 vs 27) $p < 0.001$ • Waste-separation bins by ○ Older (30.6 vs 27.5) $p = 0.008$
Actual Behaviours at home	• Save-energy F 59% (+13%) $p = 0.013$ • Reduce-waste F 49% (+15%) $p = 0.003$ • No-food-waste F 58% (+18%) $p = 0.001$ • Recycle F 77% (+20%) $p < 0.001$ • No-action-adopted M 3% (+3%) $p = 0.081$ (MS)	• save-energy by Older (32 vs 27) $p < 0.001$ • reduce-waste by Older (34 vs 26.6) $p < 0.001$ • no-food-waste by Older (31 vs 28) $p = 0.001$ • recycle by Older (31.6 vs 26) $p < 0.001$
Overall Awareness and Engagement	Females more engaged (logistic $p < 0.001$) • +82% higher odds of engagements for females vs males	• Older more engaged-Positive trend (logistic $p < 0.001$) – Each decade increases by 13% the odds of engagements ○ Using 6 age categories: ○ Positive trend confirmed ○ 45–55 more engaged ($p < 0.001$ compared to under 24) ○ 35–44 (MS when compared to under 24, $p = 0.0064$) • Over 55 (MS when compared to under 24, $p = 0.0507$ respectively)

odds of selecting a bus (odds of approximately four times higher than the corresponding odds of selecting any other mode). Conversely, considering environmental awareness as a key factor significantly decreased the odds of using a private car (by 83%) compared to any other transportation options.

Table 2 summarises the impact of various factors on travel mode choices.

Table 2. Impact of various factors on travel mode choices.

Variable	Car	Motorbike	Bus	Metro	Walking
Convenience	✓ +123% (0.001)		✓ −75% (<0.001)		
Time Saving	✓ +79.6% (0.022)	✓ +540% (<0.001)	✓ −80% (0.071)		
Cost	✓-89% (<0.001)		✓ +88% (0.072)	✓ +666% (<0.001)	
Reliable Service			✓ +130% (0.012)		**MS** ✓-74.9% (0.072)
Comfort	✓ +494% (<0.001)		✓ −82% (0.001)	✓-87% (<0.001)	
Safety	✓-83% (0.001)	−75% (0.187)		✓-75% (0.010)	
Environmental Awareness	✓ +172% (0.007)		✓ +244% (0.010)		✓ +442.8% (<0.001)

All models selected using the AIC and full stepwise method. NS effects kept by AIC are greyed out.
Each cell presents % change in Odds* and the *p*-value in parentheses; empty cells: the variable is not selected as important according to the AIC and full stepwise method; grey details: the variable was included in the AIC stepwise model but not significant at 10%.
✓ Included in final AIC stepwise model and with $p < 0.10$; MS ✓ Variable effect of marginal significance (i.e. with $0.05 < p < 0.10$).
* % Change in odds: is given by the exponent of the coefficient of the logistic regression minus 1 and multiplied by 100. It provides the percentage change of odds for the corresponding measure when the correspond level is present (for categorical variables) or when the numeric variable of interest increases by one unit with all other factors being constant.

Interestingly, the overwhelming majority of respondents (95%) expressed a commitment to engaging in environmental actions. However, while a substantial majority of respondents (59.5%) indicated that they are willing to use environmentally friendly transportation, only a small fraction (7%) acknowledged environmental considerations as a determinant in their transportation decisions. Furthermore, the data revealed that 59% of the participants are willing to utilize waste-separated bins, nearly half (45%) are open to purchasing recycled products and opting for paper cups and bottles over plastic alternatives, a smaller segment (18%) showed a preference for vegetarian options, and only few (3%) expressed a refusal to engage in any environmental actions.

In the context of environmental engagement, a gender difference is evident, with female fans reporting a significantly higher willingness to participate in environmental actions (99%) compared to their male counterparts (94%) ($p = 0.02651$). This pattern is consistently reflected across all responses (e.g. purchasing recycled products −58.3% versus 41.8% ($p = 0.0021$), using waste-separation bins −75% versus 53% ($p = 0.00051$).

Concerning the relationship between age and pro-environmental activity engagement, the findings revealed that fans who reported using environmentally friendly transportation, opting for paper cups and bottles instead of plastic and utilizing waste-separation bins tended to be older (mean age: 30–31 years) compared to those who did not (mean age: 27–28 years). Regarding environmental actions, the majority of respondents (61%) engage in recycling, 49%

report taking actions to conserve energy, 43% avoid producing food waste, 37% practice active waste reduction, and 34% make a conscious effort to purchase sustainable products.

The examination of gender differences in environmental activities reveals that females are more actively engaged in a range of eco-friendly practices, with the notable exception to purchasing sustainable products. The data indicates that women are more likely to save energy (13%), reduce waste (15%), avoid producing food waste (18%), and recycle (19%) compared to men (each difference being statistically significant at the 5% level). At the same time, statistically significant age differences were observed for actions like saving energy, reducing waste, avoiding food waste, and recycling. In all these cases, fans who reported engaging in these actions were older (mean age: 31–34 years) compared to those who did not (mean age: 25–27 years). Age differences ranged from 3.7 to 7.4 years, highlighting an underlying association between age and environmentally conscious behaviour.

Results of cluster analysis

Following the initial analysis of our data, we implemented cluster analysis for a group of respondents based on their reported environmental engagement and actions. These environmental engagement and action responses were transformed into dummy variables for the analysis. To ensure interpretability, a three-cluster solution was selected. Hierarchical clustering was used with Ward's method (minimization of the distance from the cluster centroid) using the Hamming distance for binary data. Three clusters emerged from this analysis, as they can be seen in Table 3 below.

Table 3. Results of cluster analysis.

Characteristic	Cluster 1: Environmentally Eager Fans	Cluster 2: Environmentally Conscious Fans	Cluster 3: Baseline Cluster
Number of Respondents (n)	91	207	277
Percentage of Sample (%)	15.8%	36.0%	48.2%
Average Age (years)	30.0	33.4	26.1
Gender (Proportion of Women)	30.8%	24.6%	13.2%
Environmental Awareness	Highest positive response regarding awareness influencing transport choices	High engagement but lower than Cluster 1	Lowest overall level of environmental engagement and activity
Overall Engagement	Highest reported levels of engagement but lower actual actions	High participation in environmental actions	Lowest level of engagement and actions
Actual Behaviors (e.g. recycling)	Lower proportion of positive responses than Cluster 2 but higher than baseline	Highest level of reported actions	Lowest proportion of positive environmental actions
Reasons for Travel (significant only)	Awareness plays a role, but overall rate is low	Convenience, time-saving, comfort	Lower proportion of car users, higher proportion of motorbike and bus users

Cluster 3 exhibits the lowest overall level of environmental engagement and activity. Due to its size and representing almost half of the participants in the sample, we name this group as the ***baseline cluster***. Clusters 1 and 2 display contrasting characteristics. Cluster 1 exhibits the highest reported levels of environmental engagement. However, their environmental actions do not fully comply with the environmental engagement trends, with a lower proportion of positive responses than Cluster 2 but still exceeding the baseline cluster. Cluster 2 demonstrates the highest level of reported active participation in environmental actions. This observation aligns with their responses regarding environmental engagement, which is higher than the baseline group but lower than the massively enthusiastic responses of Cluster 1.

Based on these characteristics, Cluster 2 was named the '***Environmentally Conscious Fans***', reflecting their strong reported actions, and Cluster 1 was called the '***Environmentally Eager Fans***', highlighting their high engagement but with a gap concerning the translation of that engagement into actual environmental actions.

Discussion

Through this study, what emerges is the need for us to appreciate that not all sport event goers share the same intentions and behaviours with regard to environmental sustainability. By shedding additional light on patterns in their views and actions, as well as offering a segmentation profiling based on them, this study illustrates that the differences among individuals need to be understood first, before environmental sustainability campaigns are to be created and promoted. With this, we argue that while sport can assist in marketing environmental sustainability in and through it, such efforts should be tailored based on a comprehensive understanding of the audience they are directed to. Unlike previous studies that examined campaign effectiveness, this study took a step back and focused on the fundamental stages that need to occur before campaigns are created, in order for their effectiveness to increase.

In regard to event attendees' behaviour, our findings regarding car usage to travel to the stadium (55%), show a slightly higher percentage when compared to previous research in Central Europe which reported 42,4% (Herold et al., 2024) and 36.1% (Loewen & Wicker, 2021), but slightly lower compared to the UK context, 64% (Scottish Football Association, Scottish Premier League, Scottish Football League, & Supporters Direct Scotland, 2013) and 67,5% (Dosumu et al., 2017). Additionally, the gender differences which appeared in our study, with female fans demonstrating a higher frequency of car usage (64.2%) compared to male fans (52.9%), were also apparent in the study of Herold et al. (2024) who found 37.3% versus 31.4%, respectively. Last but not least, our findings suggested a tendency of older fans to favour the use of private cars instead of public transportation, which also aligns with the findings of Herold et al. (2024) who found an increase in the likelihood to use a car depending on the age group, peaking at 35–44 age. When it comes to the determinants of the reasons of travel choice, our study is aligned with previous research which identified convenience (McCullough, 2013), comfort and cost (Loewen & Wicker, 2021) and time efficiency (Dosumu et al., 2017) as the main determinants of fans' transportation choices.

In this case, and in contrast with previous research on the matter, we could argue that while the value-action exists among fans, it manifests in different ways among the wider fanbase. As such, we argue that any marketing campaigns to promote sustainability through sport should be designed by first taking these diverse manifestations or forms of the value action gap into consideration.

Even though environmental considerations were not reported when discussing individuals' rationale behind their choice of transport, for those who considered the environmental impact of their travel, significantly higher odds were noted in them selecting the use of public transport, such as metro or bus, and opting out from using a private car. This would suggest that while environmental considerations might not be at the forefront of their wider concerns when selecting their preferred mode of transport, once people consider them, their actions can change dramatically. This, in turn, could point to the need to establish environmental concerns as a primary consideration, thus inviting for more frequent and concentrated promotional efforts on the carbon footprint of individuals' transport choices and potential alternatives.

As the results of the study showcased, the value-action gap identified in previous studies (Blake, 1999; Diekmann & Preisendörfer, 2003; Loewen & Wicker, 2021) was also evident in the sport event goers examined in our study. While the vast majority of them argued that they are committed to adopting environmentally friendly behaviours, their reported behaviours appear to deviate from their intentions. While delving deeper into why such a gap might exist, results showed that women reported a higher willingness to engage with pro-environmental sustainability actions, when compared to men, suggesting that maybe additional attention should be turned to men when environmental sustainability campaigns are being designed to make them appreciate the value of such actions. At the same time, future campaigns and policies assisting people who want to actually act on their intentions, and thus close the value-action gap, should be better tailored for a female audience that appears to be already more interested in doing so, by potentially providing them with additional incentives to do so. Regarding age, as older fans were found to be eager to engage in environmentally sustainable behaviours, we could once again argue that targeting older demographics for environmental campaigns and corresponding policies can leverage their existing commitment to sustainability. Simultaneously, new tailored campaigns could be created to increase awareness in younger audiences while potentially utilising different media that are more attractive to them, such as TikTok and Instagram.

Moving beyond this, however, to the authors' knowledge, this is the first study attempting to create fan segments based on sport event attendees' reported environmental intentions and actions, identifying three main clusters: the baseline (low overall level of environmental engagement), the Environmentally Conscious Fans, and the Environmentally Eager Fans. It is from this cluster analysis that the theoretical and practical implications of this study emerge. Loewen and Wicker (2021) created clusters, based on fans' travel behaviour, namely 'devoted travellers', 'home fans', and 'casual visitors', and highlighted that devoted travellers were significantly younger showed significantly lower levels of environmental consciousness, had the highest share of car use and the lowest of public transport, and eventually, the highest emission levels compared to the other clusters. Similarly, our study showed that the 'baseline cluster' showing the lowest overall level of environmental engagement was composed of

younger fans (26.1 years) and had the smallest proportion of women (13.2%) compared to the other clusters. However, this particular group exhibited the lowest proportion of car users compared to the other clusters, while the 'environmentally active fans' reported the highest proportion of respondents prioritizing convenience, time-saving, and comfort when selecting their mode of transportation. At the same time, the 'environmentally eager fans' reported the highest positive response (though with a low overall rate) regarding environmental awareness playing a role in their transportation selection. These discrepancies might indicate that environmental awareness and engagement do not necessarily translate to environmentally friendly transportation choices, a finding that can be attributed to contextual factors that have yet to be identified but might be nonetheless crucial into why the value action gap exists.

Regarding the results of the cluster analysis, we are presented with three segments of our total participants. The baseline cluster, the one capturing almost half of the participants, appears to be the one which is less engaged and active in terms of environmental sustainability. Interestingly, this group comprises mostly younger individuals who might not demonstrate environmental awareness, debunking the generally accepted perception that younger individuals are more environmentally aware compared with older ones yet chose to adopt environmentally friendly practices regarding their mode of transport when going to an event, potentially due to other reasons. This allows us to suggest that, while currently environmentally 'inactive', this group might have the greatest potential in understanding how easy adopting environmentally sustainable practices can be, drawing on the example of their choice of utilisation of public transport. If this group was thus targeted through appropriate means, such as social media, and with campaigns designed based on such messages, environmentally friendly intentions, and further actions could improve.

Environmentally conscious fans, which emerged as the second most popular segment of our participants, appear to capture those fans who are conscious but value other aspects, such as their convenience, time-saving, and comfort too much to make their interest translate into actions. This segment, representing more than one-third of sports event goers, claims that they value environmental sustainability and are indeed quite active in their engagement with some actions, but might be too comfortable to further engage with more. From a theoretical standpoint, this particular segment demonstrates mostly the value action gap, as described in the literature. This would suggest that targeting this segment would require appropriately designed strategies that incorporate marketing communication messages and potential incentives in order to counteract the 'obstacle' of convenience. Messages around the ease and benefits of environmentally friendly practices could be created and promoted, for example, while addressing their existing concerns and potentially offering incentives to counteract their inconvenience.

For the environmentally eager fans, capturing one-sixth of our overall participants, we note their wider enthusiasm and interest in environmental sustainability and yet their obstruction in actually engaging in such practices. As such, it could be argued whether these fans know how important environmental sustainability is, but do not know how to act upon it. Theoretical implications of this segment highlight the need for models of consumer behaviour that incorporate not only awareness but also the role of educational interventions in facilitating sustainable practices. It is thus recommended that marketing communications campaigns targeting this segment, could adopt a more informational

and educational tone, providing this segment with the tools needed and assisting them in closing the gap between their values and their actions. As is the case with the environmentally conscious fans, marketing communication campaigns should be created bearing in mind the demographics of the segments (above 30 years old and with about one-third of each segment identifying as women), which should be taken into account when designing the messages and identifying the means to promote them.

Limitations and further research

Despite its implications, there are of course limitations to the study. First, it involves a small sample, on a single sport event in a single country. Although this particular context is rather underexplored, with most studies thus far being focused on Central and Northern European, developed countries, it follows that more and also longitudinal data are needed to explore the clusters that are suggested in the study while focusing on similar or less similar contexts in other under-researched parts of the world. At the same time, the present study uses quantitative cross-sectional data acquired through online surveys, which are subject to potential bias and social desirability risks. As such, future research could explore this topic utilising different data and methods, including experimental designs and focus groups, in order to better explore the rationale behind sport event attendees' environmental behaviour and their potential responses to appropriate marketing communication campaigns, potentially utilising the cluster analysis introduced through this study.

Disclosure statement

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